Fault-Onset Labeling Bounds Reported Accuracy in the Tennessee Eastman Benchmark: A Leakage-Controlled Study of Six Classical Classifiers

Abstract

Fault diagnosis in the Tennessee Eastman Process (TEP) is a standard benchmark for data-driven process monitoring, yet published results are difficult to compare. We present a leakage-controlled benchmark of six classical classifiers: logistic regression, decision tree, random forest, histogram-based gradient boosting, linear support vector machine and XGBoost. Data were partitioned by complete simulation runs, standardization was fitted on training data only, hyperparameters were selected on a separate validation partition, variability was measured over five seeds, and the frozen model was evaluated once on the complete official test partition of 10.08 million observations. XGBoost ranked first in repeated validation (macro-F1 0.7683 ± 0.0017) and reached a test macro-F1 of 0.7200, accuracy of 0.6754 and MCC of 0.6621 under the labels distributed with the dataset.

These figures are bounded by the benchmark itself. Each fault is injected 160 samples into every 960-sample test run, so literal labeling caps the recall of every fault class at 0.8333; faults 6 and 7 attain 0.8333 and 0.8332, the ceiling itself. Rescoring the same frozen model over the post-onset segment, without refitting, raises macro-F1 to 0.8054, accuracy to 0.7949 and MCC to 0.7852, and the apparent validation-to-test generalization gap of $+0.0483$ becomes -0.0125 under a symmetric comparison. Across seven classifiers the relative accuracy gain lies between 0.142 and 0.150 against a structural bound of 0.1587, so the effect is a property of the labeling convention rather than of any estimator. Normal operation and faults 3, 9 and 15 remain mutually confounded after the correction, consistent with their known weak observability.

Keywords: Tennessee Eastman Process, fault diagnosis, machine learning, process monitoring, reproducible benchmark, fault-onset labeling

1. Introduction

Industrial processes operate through interacting physical, chemical, and control mechanisms. A local disturbance can propagate through feedback loops, alter several measured variables, and eventually compromise product quality, equipment integrity, safety, or environmental performance. Fault detection and diagnosis (FDD) therefore constitute central elements of modern process monitoring [1, 2, 3]. Detection determines whether the process has departed from acceptable operation, while diagnosis seeks to identify the underlying condition so that operators or automated systems can respond appropriately.

Model-based diagnosis can be effective when reliable first-principles models are available. However, plant-wide systems are frequently nonlinear, high-dimensional, stochastic, and only partially observed. These properties have motivated data-driven monitoring, in which statistical or machine-learning models infer diagnostic boundaries from historical measurements [4, 2]. Classical multivariate methods such as principal-component analysis (PCA), partial least squares (PLS), and Fisher discriminant analysis established important foundations for process monitoring [5]. More recent studies increasingly employ tree ensembles, support vector machines, neural networks, and sequence models to represent nonlinear relationships and dynamic behavior.

The Tennessee Eastman Process (TEP), originally introduced by Downs and Vogel [6], remains one of the most widely used benchmarks in this field. It represents a reactor–separator–recycle chemical process with 41 measured variables, 11 manipulated variables, and a diverse set of disturbances. Its continued value arises from the coexistence of relatively distinctive faults and difficult conditions whose trajectories overlap normal operation. The expanded simulation data published by Rieth et al. [7] provide hundreds of independent runs per condition and non-overlapping random seeds for development and testing, enabling large-scale and reproducible evaluation.

Despite extensive use of the TEP, reported results are often difficult to compare. Studies may split temporally adjacent observations rather than complete runs, tune models on the test set, report only accuracy, or omit class-level behavior. Treating autocorrelated observations as independent experimental replicates can produce optimistic estimates, while a high global score can hide failures in specific operating conditions. These issues are especially relevant when the normal state is confused with a fault, because false alarms can reduce operator confidence and generate unnecessary interven-

tions.

This study addresses these limitations through a leakage-controlled benchmark of six classical supervised classifiers: logistic regression, decision tree, random forest, histogram-based gradient boosting, linear support vector machine (SVM), and XGBoost. The study asks five questions:

1. Which classical model provides the strongest macro-averaged diagnostic performance under repeated validation?
2. Does the selected model maintain its performance on a complete, independent official test partition?
3. Which operating conditions account for the remaining diagnostic errors?
4. How much of the reported error is attributable to the fault-onset labeling convention rather than to the classifier?
5. Which model offers the best compromise between diagnostic performance and computational cost?

The main contribution is an auditable protocol that separates model selection from final testing, preserves complete simulation runs, quantifies variability over five seeds, reports confidence intervals, and examines precision, recall, F1, and confusion patterns for all 21 classes. Rather than presenting a single headline score, the analysis identifies a compact ambiguity group that defines the practical limit of the selected classifier.

A second contribution is methodological and, in our view, of wider consequence for the TEP literature. The expanded dataset injects each fault after a fixed warm-up interval, so a fraction of every faulty run is nominally normal yet carries the fault label. This fraction differs between development runs (4%) and official test runs (16.7%). We quantify how much of the reported error and of the apparent generalization gap is produced by this convention alone, and we report every headline metric under both the literal labeling and a post-onset labeling. To our knowledge this decomposition is not routinely reported, which may account for part of the dispersion among published TEP results.

2. Related Work

2.1. Data-driven process monitoring

Data-driven FDD spans latent-variable modeling, classification, clustering, change detection, and temporal modeling. The review by Venkatasubramanian et al. [1] established a broad taxonomy covering quantitative models,

qualitative models, and process-history-based methods. Chiang et al. [5, 4] showed how PCA, discriminant PLS, and Fisher discriminant analysis could support chemical-process diagnosis. Later reviews emphasized that industrial data may be nonlinear, dynamic, nonstationary, multimodal, and autocorrelated [2, 3]. These properties motivate flexible nonlinear classifiers, but also demand careful validation.

Multivariate statistical monitoring of the TEP has a long history. Raich and Çinar combined statistical process monitoring with disturbance diagnosis in multivariable continuous processes [25], and multiscale extensions using wavelet decompositions were introduced to separate phenomena occurring at different time scales [27]. A broader survey of plant-wide disturbance detection is given by Thornhill and Horch [30].

A substantial line of work addresses the serial correlation of process data directly. Dynamic PCA augments the data matrix with time-lagged variables to capture time-varying structure [40]; canonical variate analysis models the dynamics explicitly and was compared against PCA and DPCA on the TEP simulator by Russell et al. [41], later extended with kernel density estimation for nonlinear dynamics [42]. These methods are relevant here for a reason beyond their performance: they treat the temporal structure of a run as the object of analysis, whereas an observation-level classifier treats each row independently. The labeling question examined in Section 3.2 arises precisely at that boundary.

2.2. Classical machine learning for TEP diagnosis

The families evaluated here are standard: classification and regression trees [8], random forests [9], gradient boosting [10] and its regularized histogram-based formulation in XGBoost [11], together with multinomial logistic regression and margin-based linear classification [12] as controls that test whether approximately linear decision surfaces suffice. All implementations in this study were based on scikit-learn [13], except XGBoost, which used its reference implementation [11].

Support vector machines have been applied to the TEP in several forms. Chiang et al. combined Fisher discriminant analysis with support vector machines for fault diagnosis [28]; Kulkarni et al. incorporated process knowledge into the kernel to detect TEP faults [29]; and kernel extensions of PCA and independent component analysis were compared against SVM on the same benchmark [31]. These studies use kernels that the present work could not afford at the sample size considered here, which is the reason the linear

surrogate of Section 3.4 should be read as a lower bound rather than as the best attainable margin classifier.

Deep architectures have been applied extensively to the same benchmark. Chadha and Schwung compared feedforward and convolutional networks for TEP fault detection [36]; hybrid designs combining support vector machines with latent-variable preprocessing [39], hierarchical LSTM structures [37], and metaheuristically optimized deep models [38] have all reported strong results. Most directly comparable to the present study, Lomov et al. [35] evaluated recurrent, attention-based and temporal convolutional architectures on the same expanded TEP dataset used here. The present work does not propose a competing architecture; it examines a property of the evaluation protocol that applies to all of them.

2.3. Evaluation protocols and leakage

Concern about evaluation protocol is not specific to process monitoring. In a survey across disciplines that have adopted machine learning, Kapoor and Narayanan identified leakage in seventeen fields, affecting at least 294 publications and in several cases producing conclusions that did not survive correction [34]. Their taxonomy distinguishes eight mechanisms, two of which bear directly on the present benchmark: temporal leakage, where information from the future informs a prediction about the past, and non-independence between training and evaluation samples, where records that share a common origin are distributed across partitions and treated as independent replicates.

Both are latent in observation-level TEP classification. Consecutive samples within a simulation run are strongly autocorrelated, so a split performed at the observation level places near-duplicate records on both sides of the partition. The run-level protocol of Section 3.3 is designed to avoid this. The labeling question analyzed in this paper is related but distinct: it concerns not how records are divided, but whether the label attached to a record describes the state of the plant at that instant.

2.4. Evaluation beyond accuracy

Accuracy can be misleading when performance varies substantially among classes. Macro-F1 gives equal importance to every class by averaging class-specific F1 scores, while weighted-F1 weights them by support. Balanced accuracy averages class recalls. MCC summarizes agreement between predictions and true labels and remains informative for multiclass problems [14]. Confusion matrices are indispensable because they reveal whether errors are

Table 1: Treatment of the pre-onset window in observation-level studies on the Tennessee Eastman benchmark. Three studies on the same simulator adopt three distinct conventions: relabeling the pre-onset segment as normal, retaining it under the fault label, and not addressing it.

Study	Task	Methods	Pre-onset handling	Scored as fault
Yin et al. [32]	binary detection	SVM, PLS	stated, relabeled normal	800
Sun et al. [33]	detection + identif.	Bayesian RNN	stated, retained	960
Agarwal et al. [37]	21-class diagnosis	LSTM-SAE	not mentioned	960
This work	21-class diagnosis	six classical	stated, both protocols	960 / 800

diffuse or concentrated in specific class pairs. The present study therefore uses macro-F1 as the primary model-selection criterion and treats the other metrics as complementary evidence.

2.5. Reported treatment of the fault-onset window

The analysis reported in this paper is only consequential if the treatment of the onset window varies, or goes unstated, across published work on the same benchmark. We therefore examined the data-description and experimental-protocol sections of studies for which the full text was available, recording the diagnostic task, the stated treatment of the pre-onset segment, and the number of test samples scored against the fault label. The purpose is illustrative rather than exhaustive. Three studies on the same simulator adopt three different conventions, which is sufficient to establish that the choice is not standardized; Table 1 reports them.

The question applies to both distributions of the benchmark. In the original data of Downs and Vogel as distributed by Chiang et al. [6, 4], each condition is simulated for 24 hours in training and 48 hours in testing at a three-minute interval, giving 480 and 960 samples, with the fault injected after one hour and eight hours respectively. The pre-onset fractions are therefore 4.2% and 16.7%, essentially identical to those of the expanded dataset analyzed here. The artifact is thus a property of the benchmark design rather than of any particular redistribution of it, and the results of Section 4.5 bound what can be reported on either version.

Yin et al. [32] address the window directly. Their data description states that fault information emerges eight hours after start-up, so that the first 160 test samples appear normal while the remaining 800 are genuinely faulty, and their discussion of the classification output is explicit about the consequence: the target label is -1 over the first 160 samples and $+1$ from the 161st

onward. The pre-onset segment is thus relabeled as normal rather than scored against the fault, and their fault-detection rate is computed over 800 samples.

Sun et al. [33] state the same structure – the simulator runs 160 samples in the normal state before the disturbance is introduced, then continues for 800 further samples – but evaluate over all 960. Awareness of the injection instant is therefore not sufficient by itself; what determines comparability is what is done with the segment.

Agarwal et al. [37] describe their data as 1440 samples per class, of which the first 480 are used for calibration and the remaining 960 for testing, and identify the source distribution explicitly. The instant of fault injection is not mentioned anywhere in the data description or the experimental protocol. The consequence is not incidental to that study's argument: its stated objective is to improve the diagnosis of the incipient faults 3, 9 and 15, to which end pseudo-random binary signals are actively injected into the plant. Under literal labeling, part of the difficulty that motivates such an intervention is contributed by observations in which no fault is yet present.

The pattern across the three is not one of carelessness, and it admits a structural explanation. In binary detection the pre-onset segment has a natural destination: the normal class already exists in the problem, so relabeling those observations is the obvious treatment, and the false-alarm rate is in fact defined over precisely that segment. The convention is therefore built into the evaluation. In multiclass diagnosis the same move is no longer neutral, because assigning pre-onset observations of every faulty run to class 0 alters the prior of the normal condition and, in the present data, would add 1.6 million observations to a class that already competes with faults 3, 9 and 15. Faced with that, the simplest course is to label each run by its fault number throughout, and the distinction disappears without ever being posed as a choice. What is lost in the transition from detection to diagnosis is not information about the data but a convention for handling it.

The practical consequence is that reported figures are not commensurable across the three treatments. For a single frozen model the difference between literal and post-onset labeling is 0.0854 in macro-F1 and 0.1231 in MCC (Section 4.5), larger than the margins that typically separate competing methods on this benchmark. We do not claim a systematic review here; the point is that three conventions coexist, that the choice is seldom identified as such, and that it should be reported alongside the unit of partitioning.

3. Materials and Methods

3.1. Tennessee Eastman Process data

The TEP simulates a plant-wide reactor–separator–recycle system and includes 41 process measurements (XMEAS) and 11 manipulated variables (XMV), totaling 52 predictors [6]. The original problem statement specifies the plant but not a control structure; the simulations distributed as benchmark data use the plant-wide control scheme of Lyman and Georgakis [26], so the recorded trajectories reflect both the process dynamics and the closed loops acting on them. The classification task contained 21 labels: class 0 represented fault-free operation and classes 1–20 represented the predefined fault conditions. The CSV distribution used in this work is a format conversion of the expanded TEP simulation data of Rieth et al. [7, 15].

The four source files contained 15,330,000 observations and 55 columns: the 52 process variables plus the identifiers `faultNumber`, `simulationRun`, and `sample`. Development runs contained 500 observations each. Official test runs contained 960 observations each. Every class had 500 official test runs, yielding

$$21 \times 500 \times 960 = 10{,}080{,}000 \tag{1}$$

test observations. Each class therefore contributed 480,000 observations to the final test.

3.2. Fault-onset windows and labeling protocols

The expanded dataset is generated by simulating each condition for 25 hours (development) or 48 hours (official test) with a sampling interval of three minutes, and by injecting the fault after a fixed warm-up interval: one hour in faulty development runs and eight hours in faulty test runs [7, 16, 17]. In sample indices, the fault is active from sample 21 in development runs and from sample 161 in official test runs. Fault-free runs contain no injection.

Consequently, the labels distributed with the dataset are only conditionally correct. In a faulty development run, samples 1–20 (4.0% of the run) describe nominal operation while carrying a fault label; in a faulty official test run, samples 1–160 (16.7% of the run) do so. Two consequences follow directly and are, to our knowledge, seldom made explicit.

First, under an observation-level protocol with literal labels, the recall of every fault class is bounded above by

$$R_c^{\max} = \frac{960 - 160}{960} = 0.8333, \qquad c = 1, \ldots, 20, \tag{2}$$

whenever the classifier assigns pre-onset observations to some class other than c. No such ceiling applies to class 0, whose test runs contain no injection. Second, because the mislabeled fraction differs between development (4.0%) and test (16.7%), a metric computed on the two partitions is not measuring the same quantity, and part of any observed validation-to-test degradation is attributable to this asymmetry rather than to generalization.

We therefore report all headline metrics under two labeling protocols:

- **Literal labeling** (L): every observation retains the label distributed with the dataset. This is the protocol implicitly adopted by most published observation-level TEP results and is retained here for comparability.

- **Post-onset labeling** (P): pre-onset observations of faulty runs are excluded *at evaluation time*, so that each scored observation carries a label consistent with the physical state of the plant. The model is not refitted: protocol P rescores the predictions of the same frozen model that was fitted under protocol L, including its 4% of contaminated training observations. The comparison L versus P therefore isolates the labeling convention alone, and the reported post-onset figures are conservative, since a model additionally trained without the contaminated observations could only do better.

A third variant, in which pre-onset observations are excluded from fitting as well, was not evaluated here; it would confound the labeling effect with a change in the training distribution and is left for future work.

Protocol L is used for model selection, so that the frozen-model decision is unaffected by the present analysis. Protocol P is reported alongside it in Section 4.5. A third option, relabeling pre-onset observations as class 0, was not adopted because it alters the class prior of the normal condition and would confound the analysis of the ambiguity group.

3.3. Leakage-controlled partitioning and preprocessing

The original development runs were divided at the run level into training and validation partitions using a fixed random seed and a stratified 80/20 split within each source and class. This produced 8,400 training runs (400

per class), 2,100 validation runs (100 per class), and 10,500 official test runs (500 per class). No run was assigned to more than one partition.

All 52 predictors were standardized using

$$z_{ij} = \frac{x_{ij} - \mu_j^{(\text{train})}}{\sigma_j^{(\text{train})}}, \tag{3}$$

where $\mu_j^{(\text{train})}$ and $\sigma_j^{(\text{train})}$ were estimated exclusively from the 4.2 million training observations. The fitted transformation was then applied unchanged to validation and test data. This procedure prevented information from the validation or test partitions from influencing preprocessing.

3.4. Candidate classifiers

Six model families were compared. Hyperparameter tuning evaluated three candidates per family, producing 18 configurations. The selected configurations were subsequently frozen:

- **Logistic regression**: $C = 1.0$, maximum 1,000 iterations;
- **Decision tree**: unrestricted depth and minimum leaf size 1;
- **Random forest**: 400 trees, unrestricted depth, 50% of features considered per split, and minimum leaf size 1;
- **HistGradientBoosting**: learning rate 0.05, 350 boosting iterations, and at most 31 leaf nodes;
- **Linear SVM**: stochastic-gradient hinge-loss classifier with $\alpha = 10^{-5}$, maximum 2,000 iterations, and tolerance 10^{-3};
- **XGBoost**: 250 trees, maximum depth 6, learning rate 0.1, row subsampling 0.8, column subsampling 0.8, multiclass probabilistic objective, and histogram tree construction.

The tuning stage ranked configurations by macro-F1 and used MCC as a secondary criterion. The official test partition was not accepted as an input by either the tuning or repeated-validation programs.

The linear SVM was implemented as a stochastic-gradient surrogate rather than as an exact margin solver, because quadratic-programming solvers were not tractable at this sample size. Its score should therefore be read as a lower

bound on linear-margin performance, and the evidence for nonlinearity reported in Section 5.1 rests primarily on the multinomial logistic regression, which was optimized to convergence. Kernel SVMs were not evaluated for the same computational reason.

3.5. Repeated validation

After selecting the best configuration for each model family, variability was measured across seeds 2026–2030. For each seed, 40 complete runs per class were sampled from training and independently from validation. Thus, every repetition used 420,000 training and 420,000 validation observations while retaining the full 500-observation trajectories of all selected runs.

For each metric, the mean, sample standard deviation, and 95% confidence interval were calculated across the five repetitions. With $n = 5$, the interval was

$$\bar{x} \pm t_{0.975,4} \frac{s}{\sqrt{5}}. \tag{4}$$

Because only five seeds were used, these intervals quantify observed seed-to-seed variability but should not be interpreted as definitive evidence of pairwise statistical superiority. In particular, overlapping intervals for the two leading models were interpreted conservatively.

In addition to the interval estimates, the six model families were compared with the Friedman rank test over the five seeds treated as blocks, followed by the Nemenyi post-hoc procedure [19]. Pairwise comparisons used the Wilcoxon signed-rank test with Holm correction. The critical difference of the Nemenyi procedure was computed as $CD = q_\alpha \sqrt{k(k+1)/6N}$ with $k = 6$ models and $N = 5$ blocks. Results are reported in Section 4.1. With five blocks these procedures have low power by construction, and they are reported to document that the prespecified ranking rule was not substituted for an inferential check, not to establish pairwise superiority.

Because a single frozen evaluation cannot express sampling uncertainty, the official-test metrics were additionally accompanied by 95% intervals obtained by resampling complete runs with replacement (1,000 bootstrap replicates over the 10,500 test runs). The run, not the observation, is the resampling unit, consistent with the argument that temporally adjacent observations are not independent replicates.

3.6. Frozen final evaluation

XGBoost candidate 1 was selected because it achieved the highest repeated-validation macro-F1. The final seed was fixed at 2026 before test evaluation. The final development set combined 40 complete runs per class from training and 40 from validation, totaling 840,000 observations. The frozen model was then evaluated once on all 10,080,000 official test observations, without subsampling or post-test adjustment.

Two earlier executions aborted during structural validation, before any model fitting or metric computation, because of an incorrect run-length assertion; only that assertion was changed, and the full incident log is available in the repository.

The choice of 840,000 development observations, rather than the full 5,250,000 available, was imposed by the computational budget. To bound the effect of that choice, the frozen configuration was refitted on $\{10, 20, 40, 80, 160\}$ complete runs per class and scored on the full validation partition, sampling at the run level throughout. Results are reported in Section 4.2.

3.7. Performance measures and reproducibility

The reported measures were accuracy, balanced accuracy, precision, recall, class-specific F1, macro-F1, weighted-F1, MCC, confusion matrix, training time, inference time, inference time per observation, and serialized model size. Zero divisions in class-specific measures were assigned zero. Data and output integrity were monitored with SHA-256 hashes and manifests. The final execution used Python 3.12.13, scikit-learn 1.9.0, and XGBoost 3.4.0 on Linux.

4. Results

4.1. Model selection under repeated validation

Table 2 summarizes the five-seed validation results. XGBoost ranked first with macro-F1 of 0.7683 ± 0.0017 and MCC of 0.7426 ± 0.0018. HistGradientBoosting was a close second, with macro-F1 of 0.7643 ± 0.0034. Random forest achieved 0.7398, followed by decision tree at 0.6714. The two linear baselines performed substantially worse: logistic regression achieved 0.4685 and linear SVM achieved 0.4438.

The Friedman test over the five seeds rejected the hypothesis of equal performance among the six families ($\chi^2_5 = 24.54$, $p = 1.71 \times 10^{-4}$), with a

Table 2: Repeated-validation performance over five seeds. Macro-F1 was the prespecified primary selection criterion.

Rank	Model	Macro-F1 (mean ± SD)	95% CI	MCC (mean ± SD)
1	XGBoost	0.7683 ± 0.0017	[0.7661, 0.7705]	0.7426 ± 0.0018
2	HistGradientBoosting	0.7643 ± 0.0034	[0.7601, 0.7686]	0.7391 ± 0.0045
3	Random Forest	0.7398 ± 0.0021	[0.7372, 0.7424]	0.7159 ± 0.0021
4	Decision Tree	0.6714 ± 0.0013	[0.6699, 0.6730]	0.6537 ± 0.0015
5	Logistic Regression	0.4685 ± 0.0074	[0.4592, 0.4777]	0.4567 ± 0.0073
6	Linear SVM	0.4438 ± 0.0099	[0.4314, 0.4561]	0.4377 ± 0.0109

Kendall coefficient of concordance of $W = 0.982$. The ordering is thus almost perfectly consistent from seed to seed: XGBoost obtained the best macro-F1 in four of the five repetitions, and the mean ranks were 1.2 (XGBoost), 1.8 (HistGradientBoosting), 3.0 (random forest), 4.0 (decision tree), 5.0 (logistic regression) and 6.0 (linear SVM).

Post-hoc resolution is nevertheless limited. With $N = 5$ blocks the Nemenyi critical difference is 3.372 mean ranks, so only three of the fifteen pairwise comparisons are individually significant, all of them between a tree ensemble and a linear baseline. The separation between XGBoost and HistGradientBoosting is 0.6 mean ranks, far below the critical difference. The Wilcoxon signed-rank test cannot help here either: with five paired observations its smallest attainable two-sided p-value is 0.0625, so no comparison can reach $\alpha = 0.05$ even under a perfect win record, and after Holm correction none does.

These two facts are complementary rather than contradictory. The high concordance shows that the ranking is not an artifact of seed choice; the wide critical difference shows that five repetitions cannot certify small differences. Both support the conservative reading adopted here: the ordering is reproducible, the gap between the two leading models is not resolvable at this sample size, and additional seeds would be required to settle it.

The confidence intervals of XGBoost and HistGradientBoosting overlapped. Accordingly, XGBoost was selected by the prespecified macro-F1 rule, but the result is not presented as proof of a statistically significant difference between the two models. The large separation between nonlinear ensembles and linear baselines indicates that nonlinear interactions are central to this diagnostic task.

Table 3: Validation macro-F1 as a function of the training set size, in complete runs per class. Δ is the gain over the preceding row, that is, the effect of doubling the data.

Runs/class	Observations	F1 (L)	Δ	F1 (P)	Δ	Fit (s)
10	105,000	0.7538	–	0.7780	–	59.0
20	210,000	0.7591	0.0052	0.7847	0.0067	65.2
40	420,000	0.7672	0.0082	0.7930	0.0083	111.6
80	840,000	0.7756	0.0083	0.7988	0.0058	175.4
160	1,680,000	0.7806	0.0051	0.8016	0.0028	357.4

4.2. Sensitivity to the size of the development set

Table 3 reports macro-F1 on the complete validation partition as a function of the number of complete runs per class used for fitting, under both labeling protocols. All fits used the frozen configuration and 24 threads.

Two observations follow. First, the procedure reproduces the repeated-validation result: at 40 runs per class it yields a macro-F1 of 0.7672 against the 0.7683 ± 0.0017 of Table 2, a difference of 0.0011, which confirms that the curve is measured under the protocol of Section 3.5 rather than under a variant of it.

Second, the returns diminish. Doubling the data from 80 to 160 runs per class adds 0.0051 in macro-F1 under literal labeling and 0.0028 under post-onset labeling; quadrupling from the 40 runs per class used in the frozen fit adds 0.0134 and 0.0086 respectively. The frozen model is therefore not at the asymptote, and the figures reported in Section 4.3 are conservative in that sense, but the residual headroom is of the same order as the separation between the two leading model families and does not alter the ranking.

The comparison that matters for this paper is between that headroom and the effect under study. Quadrupling the training data from the size actually used gains 0.0086 in post-onset macro-F1; changing the labeling convention on a single fixed model changes macro-F1 by 0.0854, an order of magnitude more. Whatever remains to be gained from more data, it is small beside what is gained or lost by how the pre-onset window is scored.

4.3. Official test performance

After the protocol was frozen, the selected XGBoost model was fitted to 840,000 development observations and evaluated on the complete official test set. Table 4 presents the final metrics under literal labeling. Accuracy and

balanced accuracy were both 0.6754, macro-F1 and weighted-F1 were both 0.7200, and MCC was 0.6621. The equality of accuracy and balanced accuracy, and of macro-F1 and weighted-F1, follows from the exactly balanced class supports, and holds only under literal labeling; under the post-onset protocol of Section 4.5 the fault classes retain 400,000 observations while the fault-free class retains 480,000, so the two pairs of metrics separate. Macro-F1 exceeds accuracy because accuracy equals macro-recall under balanced supports, while precision exceeds recall in most classes; the harmonic mean of a high precision and a moderate recall is larger than the recall alone. The gap between macro-F1 and accuracy is therefore itself a signature of the systematic under-recall documented in Section 4.5.

Table 4: Frozen XGBoost performance on the official test set.

Metric	Value
Accuracy	0.6754
Balanced accuracy	0.6754
Macro-F1	0.7200
Weighted-F1	0.7200
MCC	0.6621
Training time (s)	170.34
Inference time (s)	56.57
Inference (ms/observation)	0.005612
Development observations	840,000
Test observations	10,080,000

Relative to repeated validation, macro-F1 decreased by 0.0483 and MCC by 0.0805. Three mechanisms could account for this difference: independent random states in the test simulations, the longer duration of test runs, and the larger fraction of mislabeled pre-onset observations. The third mechanism is both the largest and the only one that is purely a labeling artifact. Under literal labeling the attainable recall per fault class falls from $480/500 = 0.96$ in development runs to $800/960 = 0.8333$ in test runs, a ratio of 0.868, which is of the same order as the observed degradation. Section 4.5 shows that the gap does not merely shrink under post-onset labeling but changes sign: the post-onset test macro-F1 of 0.8054 exceeds the literal-label validation figure of 0.7683 by 0.0371.

That comparison alone would not be symmetric, because the validation figure was itself computed under literal labels and carries its own 4% pre-

onset contamination. The validation partition was therefore rescored under both protocols as well. Applying the selected XGBoost configuration to all 1,050,000 validation observations gave a literal-label macro-F1 of 0.7706, which agrees with the five-seed repeated-validation figure of 0.7683 to within 0.0023 and confirms that this partition is a faithful reference. Excluding the 40,000 pre-onset observations, 3.81% of the partition, raised macro-F1 to 0.7953, accuracy from 0.7556 to 0.7843, and MCC from 0.7440 to 0.7739.

The correction on the validation partition is thus +0.0246 in macro-F1, against +0.0854 on the test partition, a ratio of 0.29 that tracks the ratio of contaminated fractions (3.81% against 15.87%). Transferring the measured validation correction to the repeated-validation reference gives a post-onset validation macro-F1 of approximately 0.7929, against a post-onset test macro-F1 of 0.8054. Table 5 summarizes the decomposition.

Table 5: Validation-to-test difference in macro-F1 under both labeling protocols. The gap present under literal labeling does not survive a symmetric comparison.

Protocol	Validation	Test	Difference
Literal (L)	0.7683	0.7200	+0.0483
Post-onset (P)	0.7929	0.8054	−0.0125

Under a symmetric comparison the gap is not merely reduced but reversed, and its magnitude falls to roughly a quarter of the literal-label value, within the range of the seed-to-seed variability of Table 2. We therefore conclude that the reported generalization gap is an artifact of the labeling convention rather than evidence of degradation on unseen trajectories, and that the literal-label figure of 0.7200 understates the discriminative capacity of the frozen model by 0.0854 in macro-F1.

Two caveats attach to this conclusion. The validation rescoring used the tuned configuration fitted to the full training partition, not the five per-seed refits of the repeated-validation procedure, so the transferred correction is an estimate of the effect rather than an exact recomputation; the close agreement between 0.7706 and 0.7683 bounds the error of that substitution. Furthermore, the pre-onset destinations differ between partitions: 61.1% of pre-onset validation observations were assigned to the group $\{0, 3, 9, 15\}$ against 92.0% in the test partition, which is consistent with the development segment being only 20 samples long and overlapping the simulation transient.

Table 6: Class-wise results for the frozen XGBoost model on the official test set. Every class had support of 480,000 observations.

Class	Precision	Recall	F1	Class	Precision	Recall	F1
0	0.1481	0.2045	0.1718	11	0.8447	0.7095	0.7712
1	0.9894	0.8243	0.8993	12	0.9271	0.7669	0.8395
2	0.9896	0.8209	0.8974	13	0.9828	0.7444	0.8471
3	0.1636	0.4218	0.2357	14	0.9889	0.8187	0.8958
4	0.9477	0.8222	0.8805	15	0.1465	0.1978	0.1683
5	0.9879	0.8234	0.8982	16	0.9756	0.6681	0.7931
6	0.9973	0.8333	0.9080	17	0.9624	0.7809	0.8622
7	0.9964	0.8332	0.9075	18	0.9367	0.7760	0.8488
8	0.9665	0.7779	0.8620	19	0.7701	0.7559	0.7630
9	0.1328	0.2126	0.1635	20	0.8391	0.7029	0.7650
10	0.8066	0.6885	0.7429				

4.4. Class-specific performance

Table 6 demonstrates marked heterogeneity. Seventeen classes achieved F1 scores from 0.7429 to 0.9080. Classes 6 and 7 were strongest, with F1 of 0.9080 and 0.9075, respectively. Classes 1, 2, 5, and 14 also approached or exceeded 0.8958.

Four classes were persistently difficult: class 9 (F1=0.1635), class 15 (0.1683), normal operation (class 0; 0.1718), and class 3 (0.2357). Class 3 had the highest recall within this group (0.4218) but very low precision (0.1636), indicating systematic overprediction.

A pattern in the recall column deserves emphasis. Classes 6 and 7 reached recalls of 0.8333 and 0.8332, which coincide with the structural ceiling of Eq. (2) to within one part in ten thousand. Classes 1, 2, 4, 5 and 14 lie between 0.8187 and 0.8243, that is, at 98–99% of that ceiling. The frozen model is therefore essentially saturated on these classes once the pre-onset segment is removed, and their reported F1 values of roughly 0.88–0.91 are limited by the labeling convention rather than by the classifier.

4.5. Effect of the fault-onset window

Table 7 compares the frozen model under the two labeling protocols of Section 3.2. No refitting, retuning or reselection was performed; the same frozen model and the same predictions are scored under two label conventions.

Table 7: Frozen XGBoost under literal (L) and post-onset (P) labeling on the official test partition. Intervals are 95% run-level bootstrap intervals over 1,000 replicates.

Metric	Literal (L)	Post-onset (P)
Accuracy	0.6754 [0.6716, 0.6797]	0.7949 [0.7898, 0.8006]
Balanced accuracy	0.6754 [0.6748, 0.6760]	0.8006 [0.7998, 0.8013]
Macro-F1	0.7200 [0.7192, 0.7208]	0.8054 [0.8044, 0.8063]
MCC	0.6621 [0.6579, 0.6666]	0.7852 [0.7798, 0.7910]
Observations	10,080,000	8,480,000

Excluding the 1,600,000 pre-onset observations, 15.87% of the partition, raised macro-F1 by 0.0854, accuracy by 0.1195 and MCC by 0.1231. No model was refitted and no hyperparameter was changed.

Two diagnostics establish that this is a labeling effect rather than a favourable choice of subset. First, the frozen model assigned 16.10% of the pre-onset observations to class 0 and 75.86% to classes 3, 9 or 15, that is, 92.0% to the group of conditions that are indistinguishable from nominal operation, and only 4.20% to the fault label they nominally carry. The model therefore identifies the pre-onset segment as a period in which nothing detectable is occurring, which is correct, and literal labeling scores this as error. Second, restricting the evaluation to post-onset observations raised the recall of the 17 well-separated classes to between 0.8015 and 0.9999, the upper value corresponding to essentially complete recovery of faults 6 and 7, while the ambiguity group improved only marginally (F1 of 0.2217, 0.3238, 0.2100 and 0.2106 for classes 0, 3, 9 and 15, against 0.1718, 0.2357, 0.1635 and 0.1683 under literal labels). The ambiguity group is therefore not a labeling artifact; the ceiling on the remaining seventeen classes is.

The two phenomena are nevertheless connected. Class 3 was predicted 1,237,562 times against a true support of 480,000, an excess of 757,562 observations. A substantial part of that excess consists of pre-onset segments belonging to other faults, which the model routes to the weakly observable conditions because, at those instants, nothing has yet happened. The overprediction of class 3 is thus partly a consequence of the labeling convention and partly a consequence of genuine overlap, and only the post-onset analysis separates the two.

This distinction matters for comparability. As documented in Section 2.5, published observation-level results on the expanded TEP dataset do not con-

Table 8: Effect of post-onset labeling across model families. The last column reports the relative accuracy gain $\Delta A/A_P$, whose structural upper bound is $1-8{,}480{,}000/10{,}080{,}000 = 0.1587$.

Model	Macro-F1 (L)	Macro-F1 (P)	$\Delta A/A_P$
Decision tree	0.6330	0.7167	0.148
Extra trees	0.6965	0.7786	0.148
Random forest	0.7081	0.7912	0.149
HistGradientBoosting	0.7128	0.7972	0.149
XGBoost†	0.7200	0.8054	0.150
Logistic regression	0.4283	0.4698	0.142
Linear SVM	0.4207	0.4655	0.142

†Frozen model of Table 4, complete test partition.

sistently state whether pre-onset samples were retained, discarded, or relabeled, yet the three choices are separated by several points of macro-F1 for an identical model. Two studies reporting different scores may therefore differ in labeling convention rather than in method. We recommend that the treatment of the onset window be reported explicitly as part of the experimental protocol, in the same way that the unit of partitioning is reported.

4.6. Generality of the labeling effect across model families

If the effect is a property of the labeling convention rather than of the selected classifier, it should appear in every model evaluated on the same partition. An auxiliary experiment was therefore rescored under both protocols. It predates the frozen benchmark, uses a per-class subsample of the test partition and includes an extremely randomized trees model in place of XGBoost, so only the differences between protocols are interpreted. No model was refitted.

The five tree-based models gained between 0.0821 and 0.0854 in macro-F1, a spread of 0.003 across estimators whose capacities differ by orders of magnitude, from a single unpruned tree to a regularized boosting ensemble. No overfitting mechanism produces that uniformity, because the degradation associated with overfitting scales with model capacity.

The relative accuracy gain explains the pattern. If a classifier assigns pre-onset observations to any class other than the labeled fault, its literal accuracy is its post-onset accuracy scaled by the retained fraction of the partition, so $\Delta A/A_P$ is bounded above by 0.1587 independently of the model.

The seven models fall between 0.142 and 0.150, the residual being the small fraction of pre-onset observations that match the literal label by chance. The linear baselines gain less in absolute terms, 0.0415 and 0.0448, only because they start lower, and they route far fewer pre-onset observations to the group $\{0, 3, 9, 15\}$, 20.6% and 41.9% against 65.7–70.2% for the ensembles, yet are penalized in the same proportion. Any observation-level result reported on this dataset under literal labels therefore understates the post-onset discriminative capacity of the model by a factor predictable from its own accuracy.

4.7. Structure of the remaining errors

The normalized confusion matrix in Fig. 1 shows that errors were not uniformly distributed. They concentrated in the four-class group $\{0, 3, 9, 15\}$. Among true class-9 observations, 32.39% were predicted as class 3, 20.24% as class 0, and 18.08% as class 15. Normal observations were assigned to classes 3, 9, and 15 in 32.02%, 21.51%, and 18.24% of cases, respectively. For class 15, the corresponding proportions were 30.96%, 20.53%, and 20.03% for predicted classes 3, 9, and 0.

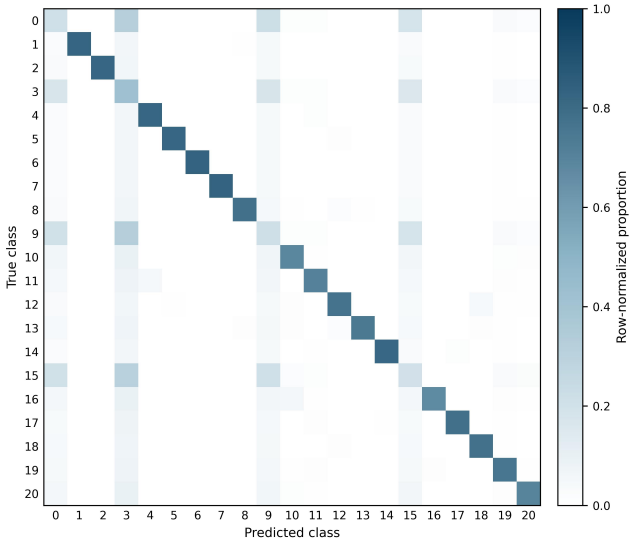

Figure 1: Row-normalized confusion matrix for the frozen XGBoost model on the official test partition.

Class 3 was predicted 1,237,562 times despite having 480,000 true observations. This excess explains its low precision and demonstrates that the

Table 9: Performance and computational cost of the six families under repeated validation (mean over five seeds, 420,000 training observations per repetition). Times measured on the environment of Section 3.7.

Model	Macro-F1	MCC	Train (s)	Inference (μs/obs)	Size (MB)
XGBoost	0.7683	0.7426	864.7	7.03	18.9
HistGradientBoosting	0.7643	0.7391	73.5	29.40	16.9
Random Forest	0.7398	0.7159	477.3	20.29	8703.8
Decision Tree	0.6714	0.6537	54.0	0.24	28.8
Logistic Regression	0.4685	0.4567	65.4	0.15	0.005
Linear SVM	0.4438	0.4377	4.1	0.16	0.006

model used class 3 as a frequent destination for ambiguous trajectories. The persistence of the same difficult group during repeated validation and official testing suggests an intrinsic overlap under an observation-level representation rather than an isolated sampling artifact.

4.8. Computational performance

Table 9 reports the cost of all six families under identical conditions during repeated validation, which is required to answer the performance–cost question and cannot be inferred from the selected model alone. Training times are reported as means over the five seeds and are not directly comparable with the 170.34 s of Table 4, which was measured during the final evaluation on twice as many observations. The degree of parallelism was an invocation-time argument of each stage rather than a fixed setting, and it was not recorded in the run manifests, so the absolute times of the two stages reflect different thread counts on a 24-core host. The comparison between families within Table 9 is unaffected, because those six runs shared a single invocation, but the cross-stage comparison is not meaningful and no conclusion is drawn from it. Recording the degree of parallelism alongside the timing is a reproducibility requirement that this benchmark did not initially meet, and it has been added to the released code.

Final XGBoost training required 170.34 seconds. Prediction over 10.08 million observations required 56.57 seconds, corresponding to 0.005612 ms per observation or approximately 178,000 observations per second. The serialized final model occupied approximately 6.8 MB. These values indicate that the selected classifier is computationally feasible for high-throughput offline analysis and potentially for online scoring, although end-to-end deployment

latency must also include data acquisition, preprocessing, communication, and decision logic.

5. Discussion

5.1. Nonlinearity and model ranking

The ranking provides strong empirical evidence that the TEP classification boundary is not adequately represented by the evaluated linear models. XGBoost, HistGradientBoosting, and random forest substantially outperformed logistic regression and linear SVM even though all models received the same standardized predictors and run-level partitions. Tree ensembles can represent threshold effects and high-order interactions without requiring an explicit feature expansion, which is consistent with the nonlinear and feedback-driven nature of the TEP.

XGBoost and HistGradientBoosting differed by 0.0040 in mean macro-F1 with overlapping intervals, so the defensible conclusion is that XGBoost won under the prespecified rule, not that it is superior under all resampling regimes. Table 9 shows that the practical choice does not favour either: HistGradientBoosting trained 11.8 times faster, while XGBoost inferred 4.2 times faster, so the decision follows from whether the deployment is retraining-bound or scoring-bound. Model size is a third and largely independent constraint: the random forest required some 460 times the XGBoost artifact for 0.0285 less macro-F1, which rules it out for embedded deployment without being visible in any predictive metric. None of these conclusions can be drawn when only the selected model is profiled.

5.2. Why global performance is insufficient

The official macro-F1 of 0.7200 summarizes useful performance, but it does not describe a uniformly reliable 21-class diagnostic system. If only macro-F1 or accuracy had been reported, the strong results for 17 classes would obscure the near-failure of normal operation and faults 3, 9, and 15. In operational terms, low precision for normal operation and extensive exchange between normal and faulty labels can increase false alarms and missed interventions.

This result illustrates why class-level precision, recall, F1, and confusion structure should accompany aggregate measures in industrial diagnosis. It also supports a distinction between *benchmark discrimination* and *deployment readiness*. The present model is a strong benchmark classifier for most

faults, but additional safeguards are necessary before it could support autonomous decisions.

5.3. Implications for method selection in industrial practice

Benchmark figures circulate as evidence of capability: a method reporting a macro-F1 of 0.80 on the TEP appears preferable to one reporting 0.72. Sections 4.5 and 4.6 show that a difference of exactly that size separates two labeling conventions applied to an identical model. The question to put to a reported figure is therefore not only which algorithm produced it, but over which observations it was computed.

The fixed injection instant of the benchmark has no counterpart in an operating plant, but the interval it stands for does. A disturbance entering a unit does not become visible in the instrumentation at the moment it occurs. It propagates through transport lag, and it is actively opposed by the control loops, whose rejection of the disturbance is precisely what conceals it: the difficulty of detecting faults inside chemical-process control loops due to feedback masking is long recognized [20]. For incipient conditions the margin is narrow to begin with, with reported amplitudes on the order of one to ten per cent of the fault-free signal [21], so the deviation may remain within normal variability for an extended period. The interval between occurrence and annunciation is itself a design quantity in alarm management, where deadbands and delay timers trade detection delay against nuisance alarms [22]. Whatever label a historical dataset attaches to observations in that interval, a model receives data carrying no signature of the condition it is expected to report, and a figure of merit computed over them measures the labeling convention rather than the diagnostic system.

The group $\{0, 3, 9, 15\}$ makes the same point in a sharper form. These conditions survive the correction of Section 4.5 because they leave no persistent signature in the 52 measured variables. For a deployment this is not a modeling deficiency to be repaired with a better classifier; it is a statement about what the installed instrumentation can observe. The literature treats this as a design problem rather than an algorithmic one: sensor location can be posed explicitly against fault-diagnostic observability criteria [23], and a comprehensive formulation combining diagnosability, reliability and sensor cost has been demonstrated on the Tennessee Eastman flowsheet itself [24]. The implication for practice is that when a condition proves undiagnosable from the available measurements, the productive response is to revisit the measurement set, not to continue tuning the classifier.

The cost results of Table 9 show that no model dominates. HistGradientBoosting trained 11.8 times faster than XGBoost for a difference of 0.0040 in macro-F1, while XGBoost inferred 4.2 times faster; the random forest required approximately 8.7 GB. Which constraint binds is a property of the installation rather than of the benchmark: a model retrained weekly on a workstation and a model scoring within the scan cycle of a controller are limited by different quantities, and neither limit is visible in a predictive metric.

The common thread is that benchmark performance and deployment readiness are distinct properties, and that the distance between them is not only a matter of model quality. It also depends on how the evaluation was constructed, on what the instrumentation can observe, and on the resources available where the model will run.

5.4. Interpretation of the ambiguity group

The identity of the difficult group is not a new finding, and it would be misleading to present it as one. Faults 3, 9 and 15 have been reported as weakly detectable since the earliest systematic studies of the TEP: monitoring statistics based on PCA, discriminant PLS and Fisher discriminant analysis report missed-detection rates close to those of nominal operation for exactly these conditions [5, 4], and later comparative studies reproduce the pattern across a wide range of data-driven statistics [18]. The contribution here is the scale and the protocol: the same group is recovered under run-level partitioning, on 10.08 million independent test observations, and it persists after the labeling artifact of Section 4.5 is removed. This strengthens the interpretation that the overlap is a property of the measured variables rather than of any particular estimator.

The mechanism, however, constrains which remedies can work. If these disturbances leave no persistent signature in the 52 measured variables, then temporal aggregation over a run reduces the variance of an estimate that is already centered on an ambiguous region: it will sharpen the decision without moving it toward the correct class. Run-level aggregation should therefore be expected to help the 17 separable classes, whose pre-onset transients are transient, far more than the group $\{0, 3, 9, 15\}$, and this asymmetry is itself a testable prediction. A hierarchical design that isolates the ambiguous group and applies a discriminator with explicit memory to classes 0, 3, 9 and 15 is the more promising direction, but its plausibility rests on whether any signature exists at all; establishing this is a question of observability, not

of classifier capacity, and it is better addressed with residual-based or first-principles analysis than with additional supervised tuning.

Both proposals arise from test-set diagnosis and must therefore be evaluated under a newly defined development/test protocol; the present official test must not be reused for tuning them.

5.5. Threats to validity

The TEP is a high-fidelity simulation, not a physical plant, so the results need not transfer to sensor drift, maintenance changes, unmodeled disturbances or evolving operating modes; and the evaluation is pointwise multiclass diagnosis, which does not measure detection delay, false alarms per run or sustained alarm logic. The uncertainty estimates rest on five seeds and characterize the implemented experiment rather than all possible partitions, and the hyperparameter search was deliberately limited to three candidates per family, so individual algorithms, particularly the linear baselines, might improve under a wider search. The linear SVM in particular was a stochastic-gradient surrogate and should not be read as the best attainable margin classifier.

Three further caveats bear on quantities reported here. The final model used 840,000 of the 5,250,000 available development observations and selection was performed at 420,000; Section 4.2 bounds the residual headroom at 0.0086 in post-onset macro-F1 under quadrupling, so the reported figures are conservative, but a ranking established at one scale need not be invariant to it. The degree of parallelism was not recorded in the run manifests, so the times in Table 9 support ordinal comparison between families, which shared one invocation, but not comparison against other stages. Finally, class support is balanced by construction, whereas industrial faults are rare; metrics, thresholds and alarm costs would require recalibration under field prevalence. Interpretation throughout focused on output behaviour and did not include feature attribution or causal analysis.

6. Conclusion

This work presented a leakage-controlled benchmark of six classical classifiers for 21-class fault diagnosis in the Tennessee Eastman Process, using run-level splitting, train-only standardization, separate tuning and validation, five-seed repetition and a single frozen test evaluation.

XGBoost ranked first in repeated validation (0.7683 ± 0.0017) and produced a test macro-F1 of 0.7200, accuracy of 0.6754 and MCC of 0.6621 over 10.08 million observations under literal labeling, rising to 0.8054, 0.7949 and 0.7852 once the pre-onset segment is excluded, with no refitting. Rescoring the validation partition under the same protocol turns an apparent generalization gap of $+0.0483$ into -0.0125. Several of the seventeen well-diagnosed classes were shown to sit at the structural recall ceiling imposed by the labeling convention rather than at the limit of the classifier, while normal operation and faults 3, 9 and 15 remain confounded after the artifact is removed.

Three conclusions follow. Nonlinear tree ensembles diagnose most TEP conditions effectively. Aggregate scores overstate practical reliability when a small group of classes dominates the remaining errors. And a benchmark result on these data is not comparable across studies unless the treatment of the fault-onset window is stated: a single frozen model differs by 0.085 in macro-F1 between two defensible conventions, more than most improvements reported between competing architectures, and the effect is common to all seven model families tested. We recommend that the treatment of the onset window be reported as part of the experimental protocol, alongside the unit of partitioning.

Future work should investigate temporal aggregation, hierarchical diagnosis, probability calibration and cost-sensitive alarm rules under a new untouched test protocol. Sequence models raise the question in a richer form: windowed labeling introduces a third category of observations, those spanning the injection instant, whose proportion depends on the windowing scheme of each study.

Data and Code Availability

The expanded Tennessee Eastman simulation data are publicly available from Harvard Dataverse [7]. The CSV conversion used in this study is available through Kaggle [15]. All code, configuration files, run-level split manifests, frozen hyperparameters, five-seed validation outputs, final predictions, class-level metrics, confusion matrices, SHA-256 data hashes, and the environment specification are available for review at an anonymized mirror, `https://anonymous.4open.science/r/SEU-LINK`. The non-anonymized repository and its archival DOI are cited in the accepted version. The analysis

of the fault-onset window can be reproduced from the deposited predictions without refitting any model.

CRediT Authorship Contribution Statement

First author: Conceptualization, Investigation, Validation, Writing – original draft, Writing – review & editing. **Second author:** Investigation, Validation, Writing – original draft, Writing – review & editing. **Third author:** Conceptualization, Methodology, Software, Formal analysis, Data curation, Writing – original draft, Visualization, Supervision, Project administration.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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